NIHR | National Institute for Health and Care Research

Reference number	NIHR208155
Programme	Research for Patient Benefit
Call	RfPB Under-represented disciplines and specialisms highlight notice: Allied Health Professionals
Lead Applicant	Dr Nicky Wilson
Host organisation	King's College Hospital NHS Foundation Trust
Research Title	Feasibility of a Community-Based Aquatic Exercise and Peer Support Intervention (Good Boost+) for Adults on Hip and Knee Replacement Waiting Lists: A Randomised Hybrid Type 1 Effectiveness-Implementation Feasibility Trial
Start Date	01/09/2025
Grant Duration (Months)	24
End Date	31/08/2027
Estimated Research Cost	£290,399.00
Estimated NHS Support & Treatment costs	£28,470.79
Research Type	Primary Research

Lead Applicant Details

Lead Applicant

Full Name	Dr Nicky Wilson	Position	Consultant Physiotherapist
ORCID iD	0000-0001-7404-7360	Institution	King's College Hospital NHS Foundation Trust

Specify your (lead applicant) role in the research
Study CI, lead and oversee management of the project. Co-ordinate and manage co-applicants, collaborators, and patient partners. Responsible for good research governance, study conduct, co-supervision of the health services researcher and comprehensive reporting of findings. Co-lead PPI/E with Barrington Foster, a Black British patient partner with lived experience of hip and knee replacement surgery. Support and mentor Barrington to establish a patient advisory group representative of the local population; act as a single point of contact for public contributors; ensure recruitment is accessible to under-served groups; and co-lead patient advisory group meetings and activities.

% FTE commitment	20

Joint Lead Applicant

Full Name	Professor Alison McGregor	Institution	Imperial College of Science, Technology and Medicine
ORCID iD	0000-0003-4672-332X	Position	Professor of Musculoskeletal Biodynamics

Justification for Joint Lead Applicant and role in research
Professor McGregor will jointly lead this project with Nicky Wilson and oversee all aspects of the research, including research governance, data management, delivery, and dissemination. Nicky Wilson is an early career researcher. Professor McGregor is an experienced and recognised researcher in the field of rehabilitation having led a range of clinical studies, including a post-operative randomised controlled trial into rehabilitation following spinal surgery. She has a strong track record of supporting and championing early career researchers. This has included a number of AHPs successfully achieving their PhD with many going on to secure academic leadership positions across the UK. Alison will support Nicky Wilson in her first large grant via fortnightly meetings to: advise on project management and risk mitigation; signpost her to opportunities, networks and resources to support her skill and capacity development; and enable her progression as a clinical researcher.

% FTE	5

Research Team Details

Co-Applicant	Organisation	Specify Co-Applicant role in research	% FTE
Ms Ruth Williams	PPI representative	Attend co-investigator and patient advisory group meetings, co-develop recruitment video, patient facing materials and plain English summaries, sense check study findings, contribute to dissemination.	1
Mr Barrington Foster	PPI representative	PPI co-lay lead, attend co-investigator meetings, co-develop recruitment video, patient facing materials and plain English summaries, sense check study findings, contribute to dissemination.	2
Professor James Galloway	King's College London	Guide and support trial set-up and delivery. Support determination of an appropriate primary outcome for a future trial, troubleshoot problems and advise on study conduct.	5
Ms Ines Reichert	King's College	Provide orthopaedic-study specific knowledge and advice. Facilitate the engagement and involvement of other orthopaedic services. Support recruitment and dissemination.	5

Co-Applicant	Organisation	Specify Co-Applicant role in research	% FTE
	London		
Dr Louise Hull	King's College London	Design and lead the qualitative implementation research, co-supervise health services researcher with Nicky Wilson, integrate implementation science methods and theories to support a future trial.	10
Professor Louise Goff	University of Leicester	Advise on and facilitate integration of EDI throughout the study. Responsible for the accessibility of trial documents and procedures to optimise reach to under-served communities.	3

COI declaration
Please declare any conflicts or potential conflicts of interest that you or your co-applicants may have in undertaking this research, including any relevant personal, non-personal and commercial interest that could be perceived as a conflict of interest.

Nicky Wilson (lead applicant) and Ben Wilkins (CEO of Good Boost) are national Musculoskeletal Clinical Champions for Versus Arthritis, a UK charity for people with arthritis. They undertook an 18-month leadership and development programme together, funded by Versus Arthritis (2018-2020). Good Boost is a social enterprise (an organisation with a social purpose), that creates Artificial Intelligence (A.I.) driven personalised exercise programmes to create more accessible options for rehabilitation and therapeutic exercise for people living with joint conditions. Nicky Wilson holds an honorary consulting role with Good Boost for the purpose of data sharing only.

Agreement to terms and conditions.	☒Confirmed

Plain English Summary of Research

Plain English summary of Reserarch

Background to this research

NHS waiting times for hip and knee replacement are long. People can wait over a year for surgery and those living in poorer areas can wait longer than others living in wealthy areas. While waiting, individuals may experience deteriorating health and social isolation, which can increase the risk of problems after surgery and NHS costs. Interventions such as exercise and psychological support, undertaken while waiting for joint replacement, can help people maintain their physical and mental health. This is called prehabilitation. However, NHS provision of prehabilitation is limited and many people don't receive it. This research aims to test a new model of prehabilitation.

Aim

We want to find out if a community-based personalised aquatic exercise and peer support programme called "Good Boost+" has the potential as a prehabilitation programme to improve physical and mental wellbeing in people waiting for a new hip or knee, whether the programme is acceptable to patients and providers, and whether or not the research to determine if it works, is possible. Good Boost+ is delivered in leisure centre swimming pools. Exercise sessions are technology supported and delivered using poolside waterproof tablet computers. Participants exercise in groups of up to eight, supervised by a physiotherapist and receive complementary post session refreshments. Participants pay £4 per session after the first four free sessions. Evidence about the difference Good Boost+ can make to physical and mental health is limited and research is needed to inform best practice.

Research plan

We propose testing the feasibility of Good Boost+ in one area in Southeast London. We will ask 48 people on NHS waiting lists for a new hip or knee to take part. A computer will be used to randomly assign participants to one of two groups. One group will receive usual care. The other group will receive Good Boost+ for six weeks in addition to usual care. All participants will complete questionnaires about their health at the start and end of the study. We will assess if people are willing to take part. We will talk to patients, NHS professionals and leisure centre staff to ask about their experiences of Good Boost+ and factors that help and hinder the programme. Patient and public members awaiting hip or knee replacement, along with those who have undergone joint replacement, helped develop this research application.

Benefits

This research will benefit patients and the NHS by testing whether the Good Boost+ programme is feasible. This information will be used to help plan a bigger study involving other locations in England if the feasibility is positive. We will share our findings with patients and organisations that support people awaiting hip or knee replacement through presentations, social media and publications.

Research Plan

Research Plan

What is the problem being addressed?

NHS waiting lists for hip and knee joint replacement are extensive. As of January 31, 2024, Trauma & Orthopaedic services in England recorded 269,610 cases where patients had been decided for hospital admission but were still awaiting it [1]. Among these, a significant proportion are individuals awaiting a new hip or knee, not uncommonly for over 12 months [2]. People in the most deprived areas are more likely to wait over a year for surgery compared to their counterparts in the least deprived areas [3].

Increased time spent waiting for a new hip or knee associates with a decrease in health-related quality of life [4, 5]. In the UK, approximately one third of individuals awaiting hip, and a quarter awaiting knee, replacement have a self-defined health status worse than death (EQ-5D score < 0) [4]. Whilst awaiting surgery, patients may experience decreased physical function and engagement in activities, including paid and unpaid work, diminished mental wellbeing, social isolation and increased opioid use [6, 7], all of which predict poor surgical outcome [8-11].

NICE guidance (NG157) recommends individuals awaiting hip or knee replacement are given advice about pre-operative rehabilitation. Notably, NICE urges research investigating prehabilitation (interventions given before surgery to optimise health and recovery), initiated at least two months before hip or knee replacement [12]. Prehabilitation aims to maintain and improve physical function and health-related quality of life pre-operatively, and may mitigate poor outcomes post-surgery. Key prehabilitation components are exercise and physical activity, psychological support and nutrition optimisation [13].

Prehabilitation provision across the UK is highly variable and limited [14]. Exercise and education are the most frequently delivered interventions, typically in hospital and community settings, although provision of resources for self-delivery is common [14]. Growing interest in digital prehabilitation to increase accessibility has led to the development of a virtual knee school for individuals awaiting knee arthroplasty, and latterly, iPREPWELL, for patients preparing for major surgery [15, 16]. However, digital solutions may lack the interpersonal interactions and social support provided by face-to-face group-based interventions, effective in facilitating adherence to physical activity and reducing loneliness [17, 18], suggesting a need for alternative effective and acceptable prehabilitation delivery models.

Currently, strong evidence on factors affecting prehabilitation uptake and participation in orthopaedic surgery patients is lacking, with limited exploration of ethnicity, age and exercise supervision [19]. However, perceived health benefits, accessible services, and support from healthcare providers, appear facilitatory [19].

These latter two factors pose problems in the NHS, prompting policy think tanks to consider alternatives to the current largely free healthcare service [20]. One such alternative is expanding charging for NHS services. However, while this may promote more responsible service use and fund additional service provision, it could exacerbate health inequalities, as individuals with low income may reduce their use of healthcare services in response to cost-sharing, albeit evidence for this primarily refers to prescription medicines and GP visits [21, 22].

Why is this research important?

Current predictions indicate a 40% increase in demand for hip and knee replacements by 2060 in the UK [23]. Simultaneously, looming funding cuts to NHS orthopaedic services casts a shadow over timely joint arthroplasty, suggesting persistent long waiting periods [24]. Exploring and evaluating prehabilitation service frameworks for patients awaiting hip and knee replacement presents a fresh avenue for advancing our understanding and support for individual health and wellbeing during the waiting period and offers potential for improved patient outcomes and reduced NHS burden.

Review of existing evidence and prior work

Systematic reviews of exercise prehabilitation interventions show moderate certainty of evidence for reduced pain, improved function, muscle strength and health-related quality of life pre-operatively in patients undergoing hip and knee arthroplasty [25-27], suggesting exercise prehabilitation may forestall deterioration or even improve patient health during the waiting period. Additionally, there is moderate certainty of evidence of reduced risk of complications and length of stay post-operatively, and some evidence of improvements in pain, function and quality of life within the first 6 months after joint arthroplasty [25-27].

In 2021, we launched a pilot community-based aquatic exercise and peer support intervention for patients with musculoskeletal (MSK) disorders under a Southeast London NHS Trust, supported by a grant from the Health Innovation Network South London. The intervention featured Artificial Intelligence (A.I.) generated personalised exercise sessions delivered via waterproof tablets, with physiotherapy staff, students and volunteers supporting a group of up to eight patients in the water. Complementary post-session tea and coffee was provided to foster patient connections.

The intervention was delivered by volunteer NHS physiotherapy staff, in partnership with Good Boost (a social enterprise that creates personalised exercise solutions), and a local leisure centre, aiming to enhance access to rehabilitation for adults with MSK disorders. Eligibility criteria included MSK symptoms and a commitment to six exercise sessions. Patients paid a nominal fee of £4 per session following a free introductory session.

The six-month pilot data showed diverse attendance demographics and high participation rates: 56% of the 34 attendees were from ethnic minority groups; 62% were from the lowest three income deciles and 71% attended ≥ six exercise sessions within three months. While musculoskeletal health clinical data collection faced challenges (some participants opted out of completing outcome measurements on the tablets), qualitative feedback from participants underscored the program's positive impact on

physical and mental well-being, social isolation, and self-management skills.

Good Boost is NHS Digital Technology Assessment Criteria (DTAC) certified and an ORCHA reviewed medical app. Good Boost technology is available in 200+ UK swimming pools. This sets the stage for a scalable opportunity to introduce an exercise and peer support prehabilitation intervention for individuals awaiting hip and knee replacement, pending its effectiveness. A recent search on the International Clinical Trials Registry Platform identified no current UK-based community aqua exercise and peer support prehabilitation trial.

Research question and objectives.

Is it feasible to deliver a randomised hybrid type 1 effectiveness-implementation trial of a community-based aquatic personalised exercise and peer support intervention called "Good Boost+" for adults on hip and knee replacement waiting lists?

Primary objective

To determine the feasibility of recruitment, retention and adherence to a trial implementing the Good Boost+ programme for individuals on waiting lists for hip and knee replacement.

Secondary objectives

1. To assess the acceptability of Good Boost+ to patients wait listed for hip and knee replacement, NHS professionals and leisure staff.
2. To determine an appropriate primary outcome measure for a definitive hybrid type 1 trial assessing the effectiveness and implementation of Good Boost+.
3. To identify determinants of implementation and develop an implementation strategy to optimise adoption and implementation of Good Boost+, to be evaluated in a multi-site hybrid type 1 trial.

Project plan

This design is informed by patients and public members, the CONSORT extension statement and guidance for conducting feasibility studies for implementation trials [28, 29]. A parallel, two arm, assessor blinded, randomised controlled feasibility study with participants allocated to either i) intervention group (Good Boost+ plus usual NHS care) or ii) control group (usual NHS care) will be employed. The study will be evaluated using mixed research methods.

Setting

Patients will be recruited from orthopaedic wait lists at King's College Hospital (KCH), Guy's & St Thomas' (GSTT) and Lewisham & Greenwich (L&G) NHS Trusts where there is a high prevalence of patients awaiting hip and knee replacement (approximately 500 patients at KCH; average wait times 5 to 6.5 months). An urban setting such as Southeast London will enable us to assess a diverse range of ethnic groups and socio-economic status.

Target population, inclusion and exclusion criteria

Patients registered on an NHS wait list for hip or knee replacement.

Inclusion criteria:

i. Aged ≥18 years
ii. Registered on a wait list for primary hip or knee replacement at KCH, GSTT or L&G with sufficient time for delivery of the intervention prior to surgery.
iii. Willingness and ability to participate in weekly water-based exercise at The Castle or Peckham Pulse swimming pools.
iv. Willingness to pay £4 per session after the first four complementary sessions.
v. Access to a smart phone or computer with internet access.
vi. Able to provide informed consent.

Exclusion criteria:

i. Individuals scheduled to have a second contra-lateral hip or knee replacement within the study period.
ii. Individuals with Inflammatory Arthritis.
iii. Individuals who self-report being advised by a healthcare professional not to exercise.

Sample size

A pragmatic representative sample in line with the INCLUDE guidance will be employed, based on predicted Good Boost+ capacity in Southwark and recommendations for feasibility trial sample size [30]. A sample of 48 allows for estimates of the primary feasibility parameter: recruitment rate with (binomial exact) 95%CI with range +/-10% for assuming 50% recruitment rate, and retention rate with maximum width +/-15%.

Data collection

Research data will be collected from participants in the intervention and control group using self-completed paper questionnaires: prior to randomisation (baseline), and at 6-weeks post randomisation. Baseline demographic and patient characteristics data will include age, sex, ethnicity, postcode, employment status, marital status, perceived loneliness, and physical activity level. Patient-reported outcome measures (PROMS) will cover function, pain, health-related quality of life, and psychosocial domains and include the: Musculoskeletal Health Questionnaire (MSK-HQ), Western Ontario and McMaster Universities Arthritis Index (WOMAC), Warwick-Edinburgh Mental Well-being Scale (WEMWBS), EQ-5D-5L, and the MOS Social Support Survey [31- 35].

Participants in the intervention arm and individuals associated with the delivery of Good Boost+, including NHS professionals and leisure staff, will complete the Theoretical Framework of Acceptability (TFA)-based questionnaire [36].

Randomisation

Consenting participants will be randomised 1:1 to the intervention or control groups using a minimisation algorithm, to include postcode and ethnicity.

Intervention arm: Good Boost+

Intervention participants will continue to receive usual NHS care, but in addition will undertake six weekly 40-minute A.I. generated personalised aqua exercise sessions (including strength and mobility exercises) at either The Castle or Peckham Pulse pools, dependent on patient preference. Participants will exercise in groups of up to eight (all people with MSK disorders). Exercise sessions will be supervised by an NHS physiotherapist and supported by physiotherapy students and volunteers. Each aqua session will be followed by complementary refreshments, to foster participant connections. After the first four free sessions, participants will pay £4 per session. In addition, participants will undertake an A.I. generated personalised land-based exercise programme, accessible via a smartphone or computer, twice weekly. These land-based exercise sessions can be undertaken alone or by joining a virtual South London Good Boost group.

Control arm: usual NHS care

Participants randomised to the control arm will continue with their usual NHS care. All participants will be under the care of a named orthopaedic consultant and will receive routine information and advice at their pre-operative clinic visit. Since usual care varies between centres, participants will complete a weekly diary to record treatments received or exercise interventions undertaken during the study period.

A priori progression criteria

Based on figures from the CHIP study (NIHR Portfolio Study ID: 43614, IRAS Number: 253604) - which recruited 12 participants per month undergoing surgery for femoral head replacement at KCH - a target of 9 patients per month (allowing for 20% drop out) within a seven-month recruitment period is feasible.

The following study traffic light progress criteria will inform conclusions about whether a definitive trial is realistic or feasible:

Green: If ≥60% of eligible patients consent and at least 9 patients are recruited per month in the initial months, we will:
- Continue with recruitment as planned.
- Increase engagement activities with the clinical team to maintain/ improve consent rates.
- Review data collection and intervention delivery processes to ensure scalability for a definitive trial.

Amber: If 30-60% of eligible patients consent and 5-8 patients are recruited per month in the initial months, we will:
- Implement targeted interventions to improve patient understanding of the study, e.g. optimising information sheets and consent processes.
- Engage with patients who declined to participate to understand their reasons and adjust recruitment strategies accordingly.
- Consider extending the recruitment period or increasing the number of sites to meet target recruitment numbers for a definitive trial.

Red: If ≤30% of eligible patients consent and ≤4 patients are recruited per month in the initial months, we will:
- Conduct an in-depth analysis to identify barriers to recruitment and consent.
- Consider revising the study design or intervention to make it more appealing or accessible to potential participants.
- Discuss the viability of continuing the study and potential modifications with stakeholders and the funder.

Feasibility:

Primary outcome: The number of eligible patients recruited to the trial.

Secondary outcomes: i) rate of retention at 6 weeks, ii) adherence rate to the Good Boost+ programme, iii) proportion of participants with missing outcome data at 6 weeks. Participant attendance at exercise sessions will be recorded via the Good Boost app. Adherence to exercise will be gathered via the app and self-completed diaries.

Acceptability

Acceptability of Good Boost+ will be assessed by:
- Self-completion of a TFA-based questionnaire at 2 and 8 weeks, administered to all participants in the intervention arm and individuals associated with the delivery of Good Boost+, including NHS professionals, managers and leisure staff.
- Qualitative in-depth semi-structured interviews with a purposive representative sample of between 10 to 15 participants in the intervention arm (including all participants who did not complete the full intervention) to explore retrospective acceptability of Good Boost+ and barriers and facilitators to participation and engagement. The interview guide will be co-developed with our patient partners and advisory group, using the TFA, and piloted.

Finally, patients who decline to participate or participants who withdraw from the study but consent to further contact (n= up to 10), will be invited to take part in a 1:1 telephone interview to explore prospective and retrospective acceptability of Good Boost+ and barriers and enablers to recruitment and retention.

Primary outcome measure selection (Lead: co-investigator Galloway)

Data from the MSK HQ, WOMAC, WEMWBS; EQ-5D-5L, MOS Social Support Survey will be analysed and performance in our cohort assessed against established questionnaire psychometric properties. We will assess unusually high variance, sensitivity to meaningful change, and vulnerability to floor and ceiling effects. Specification and justification of a clinically important target difference for the chosen primary outcome, to enable an appropriate sample size to be determined for a definitive trial, will be undertaken.

Determinants of implementation (Lead: co-investigator Hull)

To identify determinants of implementation of Good Boost+, we will apply the Consolidated Framework for Implementation Research (CFIR 2.0) [37]. The CFIR 2.0 domains of outer setting, inner setting, and implementation processes, will be the primary focus of our evaluation. We will undertake semi-structured interviews with between 10 to 15 key informants (NHS and leisure staff) to explore barriers and enablers of implementation. We will use the pre-existing CFIR interview Guide tool to develop the interview topic guide. Interview transcripts will be coded against relevant CFIR domains and constructs. Analytical summaries will be developed, and data will be displayed in tabular form to facilitate understanding about cross-site implementation.

Fidelity of Implementation

A member of the research team will undertake fidelity checks of delivery of the intervention at both pools.

Data analysis

Quantitative: Simple descriptive statistics will be used to summarise the number of patients approached, consented, and randomised, and reported in line with the CONSORT statement. Proportions of participants who attend six aqua sessions, access land-based exercises, and remain in the study at 6 weeks will be determined. Baseline participant characteristics and clinical outcome data will be summarised descriptively.

Qualitative: A hybrid approach, applying deductive and inductive analysis techniques, will be used to achieve a comprehensive understanding of the acceptability of Good Boost+ and barriers and enablers to the intervention. NVivo (QSR International) will be used to support qualitative data coding.

Inclusive involvement

We have integrated the NIHR-INCLUDE Guidance [38] to improve inclusion of under-served groups in clinical research into our study design. Our preliminary pilot data (unpublished) indicates diverse uptake and participation demographics. The lead applicant has engaged regularly with patients who attended the pilot programme to gather and address feedback about the intervention and promote inclusivity, including development and dissemination of short film clips of patients from different backgrounds talking about their programme experiences. We will co-produce a recruitment video with our patient partners and offer participants a £20 inconvenience fee for completion of study questionnaires.

Dissemination

With our patient partners and collaborators, we will design and disseminate our findings to key stakeholder groups, through newsletters and a blog, presentations, an infographic containing key conclusions and images for sharing via social media, and conferences and journals.

Patient & Public Involvement:

Continuous informal feedback about the impact of our water-based exercise and peer support programme on patients and the public has been collected since its introduction in October 2021. This has informed the design and development of this proposal, which was discussed initially with four public partners who underwent joint replacement, two of whom participated in the programme while awaiting arthroplasty.

Two diverse patient partners (male and female; Black British and White) experienced in hip and knee arthroplasty are co-applicants on this study. They were involved in developing the research design, advising on usual care categorisation, PPI costs, PROMs and recruitment strategies. Their input led to inclusion of the recruitment video.

The CI met with one patient partner and four public members from diverse geographical backgrounds awaiting or with experience of arthroplasty on 14 March 2024. They highlighted the detrimental effects of prolonged waiting and emphasised preserving function. The group endorsed a community-based aquatic exercise and peer support prehabilitation intervention and discussed nominal payment. All expressed readiness to pay for sessions but noted potential drawbacks for certain individuals. These perspectives influenced the research design, especially the introduction of payment after four sessions and the choice of outcome measures.

Going forward, we will convene a patient advisory group to work with the researchers on representative recruitment, co-design of participant facing materials, interpretation of study findings, the dissemination strategy, and plain English summaries.

Team expertise

Two patient partners with experience of waiting for joint arthroplasty, and clinical and non-clinical academics with expertise in rehabilitation, orthopaedics, rheumatology, clinical trials, implementation science and health inequalities.

Project timelines

Protocol writing, ethics, set-up: 01.09.2025-28.02.2026.

Recruitment: 01.03.2026 – 30.09.2026

Qualitative data collection: 01.10.2026 -31.12.2026

Analysis and dissemination: 01.01.2027-31.08.2027

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Uploads

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good boost

Good Boost Wellbeing Limited
82 Southwark Bridge Road
London
SE1 0AS

13th February 2024

Dear Dr. Nicky Wilson,

Good Boost is a social enterprise with a mission to improve the options available for supported self-management for people living with MSK conditions which are accessible and affordable, overcoming health inequalities. Our work to date has led the community 'MSK Hubs' programme, which has been transforming community venues to support local people in condition self management.

We are more than happy to support your research project as a project collaborator.

Best wishes,

Ben Wilkins,
CEO, Good Boost

NHS
Lewisham and Greenwich
NHS Trust

Nicky Wilson
Consultant Physiotherapist
King's College Hospital NHS Foundation Trust

8th April 2024

Dear Nicky

I am writing in support your proposed research project:

> Feasibility of a Community-Based Aquatic Exercise and Peer Support Intervention (Good Boost+) for Adults on Hip and Knee Replacement Waiting Lists: A Randomised Hybrid Type 1 Effectiveness Implementation Feasibility Study

This innovative project addresses a critical need in our community and has the potential to significantly impact the well-being and outcomes of individuals awaiting hip and knee replacement surgeries.

As you know, wait times around the country are unacceptably long and your project presents a promising opportunity to mitigate the risks associated with this waiting period and to make a meaningful difference to the lives of patients awaiting joint arthroplasty.

I am supportive of screening of our local orthopaedic waiting list to identify patients within 2-3 months of primary hip or knee replacement surgery eligible to participate in your study.

I wish you success with your application to undertake this important work.

Sincerely,

Tom White

Consultant Orthopaedic Surgeon

Lewisham and Greenwich NHS Trust